🚨 SMART EMERGENCY RESPONSE SYSTEM (SERS)

Real-Time Crisis Detection & Coordination Platform for Hospitality

Solution Challenge 2026 | GDG Hackathon

📌 1. Project Overview

The **Smart Emergency Response System (SERS)** is a prototype web-based application designed to simulate real-time emergency detection, reporting, and coordination in hospitality environments such as hotels, resorts, and airports.

The system ensures instant communication between guests, staff, and management, reducing response time and improving safety during emergencies.

🎯 2. Objectives

- Reduce emergency response time
- Eliminate communication gaps
- Enable real-time coordination
- Improve safety in hospitality environments
- Provide a scalable solution for multiple locations

💻 3. Technology Stack

Frontend

- HTML
- CSS
- JavaScript

Backend

- Python (Flask)

Database

- MySQL

Dashboard

- Web-based dashboard / Power BI

☁️ 3.1 Google Tools Integration (Hackathon Advantage)

- Firebase Authentication – User login & role management
- Firebase Firestore – Real-time scalable database
- Firebase Cloud Messaging – Instant alerts & notifications
- Google Maps Platform – Location tracking & visualization
- Gemini API – AI-based emergency classification & summaries
- Vertex AI – Future predictive analytics
- Firebase Hosting – Deployment
- Google Cloud Functions – Backend automation
- BigQuery – Data analytics

⚙️ 4. Key Features

- 🚨 SOS Emergency Button
- 🧠 Multi-Level Emergency Classification
- 📊 Real-Time Dashboard
- 👥 Smart Staff Assignment
- 📢 Mass Alert System
- 🚑 Simulated Emergency Service Notification

🚦 5. Multi-Level Emergency System

🟢 Level 1: Minor Emergency

Example: Small injury, minor issue
Action:

- Only staff notified
- No disturbance to guests

🟡 **Level 2: Moderate Emergency**

Example: Medical emergency
Action:

- Staff + Manager notified

🔴 **Level 3: Critical Emergency**

Example: Fire, major threat
Action:

- All guests alerted (alarm system)
- Staff notified
- Emergency services alerted (simulated)

🔄 **6. Application Workflow**

1. User opens application
2. Clicks SOS button
3. System captures emergency details (type, location, time)
4. Emergency is classified (Minor / Moderate / Critical)
5. Data stored in database
6. Alerts sent accordingly
7. Staff assigned automatically
8. Dashboard updated in real-time
9. Emergency handled

10. Status marked as resolved

🧩 7. System Modules

7.1 User Interface Module

- SOS button
- Emergency type selection
- Status display

7.2 Alert Processing Module

- Captures user input
- Processes emergency logic

7.3 Staff Assignment Module

- Assigns available staff
- Uses predefined logic

7.4 Dashboard Module

Displays:

- Active emergencies
- Assigned staff
- Status updates

7.5 Database Module

- Stores all emergency records
- Maintains history

🌐 8. Scalability Solution (Multi-Location Support)

🔑 Key Concept: Location ID

Each hospitality location (hotel/resort) is assigned a **unique Location ID**.

🧠 How It Works

- Emergency data stored with Location ID
- Dashboard filters data based on location
- Staff access restricted to their location

✅ Benefits

- One application for multiple hotels
- Easy maintenance
- Highly scalable
- Cost-effective

⚠️ 9. Prototype Limitations

- Real-time GPS tracking is simulated
- Notifications are not integrated with real systems
- Emergency service alerts are simulated
- Staff assignment is based on predefined logic

🚀 10. Future Enhancements

- AI-based emergency prediction
- Integration with IoT devices (fire alarms, sensors)
- Real-time GPS tracking

- Integration with police, fire stations, and hospitals
- Mobile application development

🔥 11. WOW Features (Innovation Layer)

🧠 AI Panic Detection

Detects panic behavior even if user cannot clearly report emergency.

🤫 Silent SOS Mode

Hidden emergency trigger for sensitive situations.

🎬 Crisis Replay Mode

Replay emergency timeline to analyze response efficiency.

🎤 Voice SOS

Users can report emergencies using voice in natural language.

🚚 Smart Staff Assignment (Uber-like)

Automatically assigns nearest and most suitable staff.

🔄 AI Severity Upgrade

Automatically escalates emergency level if situation worsens.

📍 Smart Indoor Location Tracking

Tracks exact location inside buildings (room-based detection).

📢 Smart Targeted Alerts

Sends alerts only to affected areas to avoid panic.

🌐 Multi-Property Command Center

Single dashboard to monitor multiple locations.

🎮 Staff Training Simulation Mode

Simulates emergency scenarios for staff training.

💰 12. Business Model

Free Tier

- Small hotels
- Startups

Paid Tier

- Large hotels
- Airports
- Hospitality chains

Revenue Streams

- Subscription plans
- Enterprise licensing
- Analytics dashboard services

🌍 13. Impact

- Faster emergency response

- Improved safety standards
- Better coordination
- Scalable crisis management system

💡 14. Key Innovation

“From SOS trigger to coordinated response in seconds.”

🏁 15. Conclusion

The **Smart Emergency Response System (SERS)** provides a practical and scalable solution for handling emergencies in hospitality environments.

By combining:

- Real-time alerts
- Smart classification
- Automated coordination

The system ensures **quick action, reduced panic, and enhanced safety**.